In [1]:
```
import pandas as pd
```

In [6]:
```
df = pd.read_excel(r"C:\Users\Suresh Kunala\Downloads\data.xlsx")
df
```

Out[6]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [7]:
```
len(df)
```

Out[7]:
```
195
```

In [8]:
```
df.shape
```

Out[8]:
```
(195, 5)
```

In [10]:
```
df.columns
```

Out[10]:
```
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup'],
      dtype='object')
```

In [11]:
```
df.isnull()
```

Out[11]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

195 rows × 5 columns

In [12]:
```
df.isna()
```

Out[12]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

195 rows × 5 columns

In [13]:
```
df.isnull().sum()
```

Out[13]:
```
CountryName      0
CountryCode      0
BirthRate        0
InternetUsers    0
IncomeGroup      0
dtype: int64
```

In [15]:
```
type(df)
```

Out[15]:
```
pandas.core.frame.DataFrame
```

In [16]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 195 entries, 0 to 194
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   CountryName    195 non-null    object
 1   CountryCode    195 non-null    object
 2   BirthRate      195 non-null    float64
 3   InternetUsers  195 non-null    float64
 4   IncomeGroup    195 non-null    object
dtypes: float64(2), object(3)
memory usage: 7.7+ KB
```

In [18]:
```
df.dtypes
```

Out[18]:
```
CountryName       object
CountryCode       object
BirthRate        float64
InternetUsers    float64
IncomeGroup       object
dtype: object
```

In [19]:
```
df.head
```

Out[19]:
```
<bound method NDFrame.head of              CountryName CountryCode  BirthRate  InternetUsers  \
0                  Aruba         ABW     10.244           78.9
1            Afghanistan         AFG     35.253            5.9
2                 Angola         AGO     45.985           19.1
3                Albania         ALB     12.877           57.2
4    United Arab Emirates         ARE     11.044           88.0
..                    ...         ...        ...            ...
190          Yemen, Rep.         YEM     32.947           20.0
191         South Africa         ZAF     20.850           46.5
192     Congo, Dem. Rep.         COD     42.394            2.2
193               Zambia         ZMB     40.471           15.4
194             Zimbabwe         ZWE     35.715           18.5

            IncomeGroup
0           High income
1            Low income
2    Upper middle income
3    Upper middle income
4           High income
..                   ...
190  Lower middle income
191  Upper middle income
192          Low income
193  Lower middle income
194          Low income

[195 rows x 5 columns]>
```

In [21]:
```
df.head(10)
```

Out[21]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9000	High income
1	Afghanistan	AFG	35.253	5.9000	Low income
2	Angola	AGO	45.985	19.1000	Upper middle income
3	Albania	ALB	12.877	57.2000	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0000	High income
5	Argentina	ARG	17.716	59.9000	High income
6	Armenia	ARM	13.308	41.9000	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.4000	High income
8	Australia	AUS	13.200	83.0000	High income
9	Austria	AUT	9.400	80.6188	High income

In [22]:

```
df.head(20)
```

Out[22]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.90000	High income
1	Afghanistan	AFG	35.253	5.90000	Low income
2	Angola	AGO	45.985	19.10000	Upper middle income
3	Albania	ALB	12.877	57.20000	Upper middle income
4	United Arab Emirates	ARE	11.044	88.00000	High income
5	Argentina	ARG	17.716	59.90000	High income
6	Armenia	ARM	13.308	41.90000	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.40000	High income
8	Australia	AUS	13.200	83.00000	High income
9	Austria	AUT	9.400	80.61880	High income
10	Azerbaijan	AZE	18.300	58.70000	Upper middle income
11	Burundi	BDI	44.151	1.30000	Low income
12	Belgium	BEL	11.200	82.17020	High income
13	Benin	BEN	36.440	4.90000	Low income
14	Burkina Faso	BFA	40.551	9.10000	Low income
15	Bangladesh	BGD	20.142	6.63000	Lower middle income
16	Bulgaria	BGR	9.200	53.06150	Upper middle income
17	Bahrain	BHR	15.040	90.00004	High income
18	Bahamas, The	BHS	15.339	72.00000	High income
19	Bosnia and Herzegovina	BIH	9.062	57.79000	Upper middle income

In [23]:
```
df.head(16)
```

Out[23]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9000	High income
1	Afghanistan	AFG	35.253	5.9000	Low income
2	Angola	AGO	45.985	19.1000	Upper middle income
3	Albania	ALB	12.877	57.2000	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0000	High income
5	Argentina	ARG	17.716	59.9000	High income
6	Armenia	ARM	13.308	41.9000	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.4000	High income
8	Australia	AUS	13.200	83.0000	High income
9	Austria	AUT	9.400	80.6188	High income
10	Azerbaijan	AZE	18.300	58.7000	Upper middle income
11	Burundi	BDI	44.151	1.3000	Low income
12	Belgium	BEL	11.200	82.1702	High income
13	Benin	BEN	36.440	4.9000	Low income
14	Burkina Faso	BFA	40.551	9.1000	Low income
15	Bangladesh	BGD	20.142	6.6300	Lower middle income

In [24]:
```
df
```

Out[24]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [25]:
```
df.tail(10)
```

Out[25]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
185	Virgin Islands (U.S.)	VIR	10.700	45.3	High income
186	Vietnam	VNM	15.537	43.9	Lower middle income
187	Vanuatu	VUT	26.739	11.3	Lower middle income
188	West Bank and Gaza	PSE	30.394	46.6	Lower middle income
189	Samoa	WSM	26.172	15.3	Lower middle income
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

In [26]:
```
df.tail(15)
```

Out[26]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
180	Uruguay	URY	14.374	57.69	High income
181	United States	USA	12.500	84.20	High income
182	Uzbekistan	UZB	22.500	38.20	Lower middle income
183	St. Vincent and the Grenadines	VCT	16.306	52.00	Upper middle income
184	Venezuela, RB	VEN	19.842	54.90	High income
185	Virgin Islands (U.S.)	VIR	10.700	45.30	High income
186	Vietnam	VNM	15.537	43.90	Lower middle income
187	Vanuatu	VUT	26.739	11.30	Lower middle income
188	West Bank and Gaza	PSE	30.394	46.60	Lower middle income
189	Samoa	WSM	26.172	15.30	Lower middle income
190	Yemen, Rep.	YEM	32.947	20.00	Lower middle income
191	South Africa	ZAF	20.850	46.50	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.20	Low income
193	Zambia	ZMB	40.471	15.40	Lower middle income
194	Zimbabwe	ZWE	35.715	18.50	Low income

INDEXING & SLICING IN PANDAS

In [27]:
```
df
```

Out[27]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [29]:
```
df[:]
```

Out[29]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [30]:
```
df[1:5]
```

Out[30]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [35]:
```
df[::5]
```

Out[35]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.900000	High income
5	Argentina	ARG	17.716	59.900000	High income
10	Azerbaijan	AZE	18.300	58.700000	Upper middle income
15	Bangladesh	BGD	20.142	6.630000	Lower middle income
20	Belarus	BLR	12.500	54.170000	Upper middle income
25	Barbados	BRB	12.188	73.000000	High income
30	Canada	CAN	10.900	85.800000	High income
35	Cameroon	CMR	37.236	6.400000	Lower middle income
40	Costa Rica	CRI	15.022	45.960000	Upper middle income
45	Germany	DEU	8.500	84.170000	High income
50	Ecuador	ECU	21.070	40.353684	Upper middle income
55	Ethiopia	ETH	32.925	1.900000	Low income
60	Gabon	GAB	30.555	9.200000	Upper middle income
65	Gambia, The	GMB	42.525	14.000000	Low income
70	Greenland	GRL	14.500	65.800000	High income
75	Honduras	HND	21.593	17.800000	Lower middle income
80	India	IND	20.291	15.100000	Lower middle income
85	Israel	ISR	21.300	70.800000	High income
90	Kazakhstan	KAZ	22.730	54.000000	Upper middle income
95	Korea, Rep.	KOR	8.600	84.770000	High income
100	Libya	LBY	21.425	16.500000	Upper middle income
105	Lithuania	LTU	10.100	68.452900	High income

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
110	Moldova	MDA	12.141	45.000000	Lower middle income
115	Mali	MLI	44.138	3.500000	Low income
120	Mozambique	MOZ	39.705	5.400000	Low income
125	Namibia	NAM	29.937	13.900000	Upper middle income
130	Netherlands	NLD	10.200	93.956400	High income
135	Pakistan	PAK	29.582	10.900000	Lower middle income
140	Poland	POL	9.600	62.849200	High income
145	Qatar	QAT	11.940	85.300000	High income
150	Sudan	SDN	33.477	22.700000	Lower middle income
155	El Salvador	SLV	17.476	23.109300	Lower middle income
160	Suriname	SUR	18.455	37.400000	Upper middle income
165	Seychelles	SYC	18.600	50.400000	High income
170	Tajikistan	TJK	30.792	16.000000	Lower middle income
175	Tunisia	TUN	19.800	43.800000	Upper middle income
180	Uruguay	URY	14.374	57.690000	High income
185	Virgin Islands (U.S.)	VIR	10.700	45.300000	High income
190	Yemen, Rep.	YEM	32.947	20.000000	Lower middle income

In [37]:
```
df[10:20:3]
```

Out[37]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
10	Azerbaijan	AZE	18.300	58.7000	Upper middle income
13	Benin	BEN	36.440	4.9000	Low income
16	Bulgaria	BGR	9.200	53.0615	Upper middle income
19	Bosnia and Herzegovina	BIH	9.062	57.7900	Upper middle income

In [38]:

```
df[10:100:5]
```

Out[38]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
10	Azerbaijan	AZE	18.300	58.700000	Upper middle income
15	Bangladesh	BGD	20.142	6.630000	Lower middle income
20	Belarus	BLR	12.500	54.170000	Upper middle income
25	Barbados	BRB	12.188	73.000000	High income
30	Canada	CAN	10.900	85.800000	High income
35	Cameroon	CMR	37.236	6.400000	Lower middle income
40	Costa Rica	CRI	15.022	45.960000	Upper middle income
45	Germany	DEU	8.500	84.170000	High income
50	Ecuador	ECU	21.070	40.353684	Upper middle income
55	Ethiopia	ETH	32.925	1.900000	Low income
60	Gabon	GAB	30.555	9.200000	Upper middle income
65	Gambia, The	GMB	42.525	14.000000	Low income
70	Greenland	GRL	14.500	65.800000	High income
75	Honduras	HND	21.593	17.800000	Lower middle income
80	India	IND	20.291	15.100000	Lower middle income
85	Israel	ISR	21.300	70.800000	High income
90	Kazakhstan	KAZ	22.730	54.000000	Upper middle income
95	Korea, Rep.	KOR	8.600	84.770000	High income

In [42]:
```
df[:50]
```

Out[42]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.90000	High income
1	Afghanistan	AFG	35.253	5.90000	Low income
2	Angola	AGO	45.985	19.10000	Upper middle income
3	Albania	ALB	12.877	57.20000	Upper middle income
4	United Arab Emirates	ARE	11.044	88.00000	High income
5	Argentina	ARG	17.716	59.90000	High income
6	Armenia	ARM	13.308	41.90000	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.40000	High income
8	Australia	AUS	13.200	83.00000	High income
9	Austria	AUT	9.400	80.61880	High income
10	Azerbaijan	AZE	18.300	58.70000	Upper middle income
11	Burundi	BDI	44.151	1.30000	Low income
12	Belgium	BEL	11.200	82.17020	High income
13	Benin	BEN	36.440	4.90000	Low income
14	Burkina Faso	BFA	40.551	9.10000	Low income
15	Bangladesh	BGD	20.142	6.63000	Lower middle income
16	Bulgaria	BGR	9.200	53.06150	Upper middle income
17	Bahrain	BHR	15.040	90.00004	High income
18	Bahamas, The	BHS	15.339	72.00000	High income
19	Bosnia and Herzegovina	BIH	9.062	57.79000	Upper middle income
20	Belarus	BLR	12.500	54.17000	Upper middle income
21	Belize	BLZ	23.092	33.60000	Upper middle income

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
22	Bermuda	BMU	10.400	95.30000	High income
23	Bolivia	BOL	24.236	36.94000	Lower middle income
24	Brazil	BRA	14.931	51.04000	Upper middle income
25	Barbados	BRB	12.188	73.00000	High income
26	Brunei Darussalam	BRN	16.405	64.50000	High income
27	Bhutan	BTN	18.134	29.90000	Lower middle income
28	Botswana	BWA	25.267	15.00000	Upper middle income
29	Central African Republic	CAF	34.076	3.50000	Low income
30	Canada	CAN	10.900	85.80000	High income
31	Switzerland	CHE	10.200	86.34000	High income
32	Chile	CHL	13.385	66.50000	High income
33	China	CHN	12.100	45.80000	Upper middle income
34	Cote d'Ivoire	CIV	37.320	8.40000	Lower middle income
35	Cameroon	CMR	37.236	6.40000	Lower middle income
36	Congo, Rep.	COG	37.011	6.60000	Lower middle income
37	Colombia	COL	16.076	51.70000	Upper middle income
38	Comoros	COM	34.326	6.50000	Low income
39	Cabo Verde	CPV	21.625	37.50000	Lower middle income
40	Costa Rica	CRI	15.022	45.96000	Upper middle income
41	Cuba	CUB	10.400	27.93000	Upper middle income
42	Cayman Islands	CYM	12.500	74.10000	High income
43	Cyprus	CYP	11.436	65.45480	High income

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
44	Czech Republic	CZE	10.200	74.11040	High income
45	Germany	DEU	8.500	84.17000	High income
46	Djibouti	DJI	25.486	9.50000	Lower middle income
47	Denmark	DNK	10.000	94.62970	High income
48	Dominican Republic	DOM	21.198	45.90000	Upper middle income
49	Algeria	DZA	24.738	16.50000	Upper middle income

In [ ]:

In [ ]:

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